Clark-Quantize: Data-Free Optimal-Scale Ternary Quantization for Generative Transformers

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Abstract

We study post-training quantization of generative transformers to ternary weights $\{-1, 0, +1\}$ at an effective 1.6–2.5 bits per weight, under the strict constraint of using *no calibration data, no gradients, and no fine-tuning*. We make a single, sharp observation: at this precision the entire quality gap is controlled by the per-group *scale*, not by error feedback, group size, or which layers are protected. The widely used `absmean` rule ($s = \text{mean}\,|w|$) is provably suboptimal and, applied to a diffusion transformer, produces pure noise. Replacing it with the closed-form mean-squared-error (MSE) optimal ternary scale—which reduces to maximizing $(\sum_{\text{top-}k} |w|)^2/k$ per group—recovers coherent, prompt-faithful image generation with 94.6% of the transformer's weights ternary and only six small layers kept at original precision. On Sana 1.6B the optimal-scale quantizer improves PSNR-to-baseline from 9.1 dB (`absmean`, indistinguishable from noise) to 12.0 dB at group size 32, entirely data-free. We then show that a self-calibrated activation-aware extension—GPTQ with a shared incoherence rotation, using only the model's own generations and no external dataset—reaches 17.0 dB at the 2.1-bit budget, i.e. near-FP16 image quality. We provide the derivation, an open implementation, and ablations.

1 Introduction

Deployment cost—memory footprint, bandwidth, and energy—now dominates the economics of generative models on edge hardware. Token (or denoising-step) latency is bound by the cost of streaming weights from memory, so reducing weight precision attacks the primary bottleneck directly. Ternary weights $\{-1, 0, +1\}$ sit near the practical floor of representation: $\log_2 3 \approx 1.585$ bits, plus a small per-group scale overhead.

Most ternary results rely on quantization-aware training or curated calibration sets. We ask a narrower question: *how good can a purely data-free ternary post-training quantizer be?* A data-free method needs no held-out corpus, no forward passes over data, and integrates trivially into any toolchain. Our finding is that the answer hinges almost entirely on one design choice—the per-group scale—and that the optimal data-free choice is a closed form that is cheap to compute and markedly better than the common heuristic.

2 Method

2.1 Group-wise ternary format

For a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ we partition each row into contiguous groups of g weights along the input dimension. Each group G stores one FP16 scale s_G and a ternary code $t_i \in \{-1, 0, +1\}$

per weight, so

$$w_i \approx s_G\, t_i, \qquad t_i \in \{-1, 0, +1\}. \tag{1}$$

Storing t_i in two bits gives $2 + 16/g$ bits/weight (e.g. 2.125 at g=128, 2.5 at g=32); a tight packing approaches $1.585 + 16/g$. The format is identical across choices of the quantizer; only (s_G, t) change.

2.2 The MSE-optimal scale

Fix a group with weights $w_1, \ldots, w_g$. A ternary assignment is determined by a *support* $S \subseteq \{1, \ldots, g\}$ (the indices set to ± 1, with sign $\mathrm{sign}(w_i)$) and a scale $s > 0$. The reconstruction error is

$$\mathcal{E}(S, s) \;=\; \sum_{i \notin S} w_i^2 \;+\; \sum_{i \in S} \big(|w_i| - s\big)^2. \tag{2}$$

Minimizing over s for fixed S gives the optimal scale $s^\star = \frac{1}{|S|} \sum_{i \in S} |w_i|$, and substituting back,

$$\min_s \mathcal{E}(S, s) \;=\; \sum_{i=1}^{g} w_i^2 \;-\; \frac{\big(\sum_{i \in S} |w_i|\big)^2}{|S|}. \tag{3}$$

Since $\sum_i w_i^2$ is constant, minimizing (3) is equivalent to

$$\max_S \; \frac{\big(\sum_{i \in S} |w_i|\big)^2}{|S|}. \tag{4}$$

For a fixed cardinality $k = |S|$, the numerator is maximized by the k largest $|w_i|$; hence the optimal support is a *top-k* set and we need only search $k \in \{1, \ldots, g\}$. Sorting $|w|$ once and scanning the prefix sums yields the global optimum in $O(g \log g)$ per group:

$$k^\star = \arg\max_k \frac{c_k^2}{k}, \quad c_k = \sum_{j=1}^{k} |w|_{(j)}, \quad s^\star = \frac{c_{k^\star}}{k^\star}, \tag{5}$$

where $|w|_{(1)} \geq |w|_{(2)} \geq \cdots$ are the sorted magnitudes. This is fully data-free.

2.3 Why `absmean` is suboptimal

The common 1.58-bit heuristic sets $s = \mathrm{mean}_i\, |w_i|$ over the whole group and rounds $t_i = \mathrm{clip}(\mathrm{round}(w_i/s), -1, 1)$, i.e. a fixed threshold at $0.5\, s$. This never solves (4): it couples the scale to the *mean* rather than to the chosen support, biasing s small so that too many mid-magnitude weights round to ± 1 at the wrong amplitude. The discrepancy is mild for a single layer but compounds across dozens of transformer blocks; on a diffusion trunk it is catastrophic (Section 3).

2.4 Data-free refinements (and why they are unnecessary)

We also implemented two further data-free options and found neither helps: **(i) Error feedback.** A sigma–delta pass accumulates each weight's rounding error into the next weight before rounding, which can flip signs and better preserves running sums. With the optimal threshold scale it slightly *raised* reconstruction error and degraded images. **(ii) Per-row gain.** A least-squares per-output-channel gain $g_i = \langle W_i, \hat{W}_i \rangle / \langle \hat{W}_i, \hat{W}_i \rangle$ is a no-op once s is MSE-optimal ($g_i \approx 1$). Both are provided for completeness and are off by default.

2.5 Layer sensitivity

A small set of layers gate every block: adaptive-norm modulation, time/text conditioning, and the final projection. They are a tiny fraction of parameters but ternarizing them destroys the output regardless of scale. We keep these at the original precision (5.4% of parameters in Sana 1.6B) and ternarize the remaining 94.6%.

2.6 Evaluation protocol

We use *fake quantization*: each weight is replaced by its dequantized ternary value $s_G t_i$ but kept in bf16, so the model runs unchanged and any change in output is attributable solely to the quantizer. This isolates the algorithm's quality impact from kernel/packing concerns.

3 Experiments

Setup. We quantize the transformer of Sana 1.6B (512 px) and generate 512×512 images with 20 flow-matching steps, guidance 4.5, and a fixed seed across three prompts. We report PSNR of each quantized model's images against the FP16 baseline (higher is closer); PSNR understates quality for the stylized outputs but ranks configurations reliably.

Method	eff. bits/wt	% ternary	PSNR vs FP16 ↑	outcome
`absmean`, g=128	2.1	94.6	9.08	noise
error feedback, g=128	2.1	94.6	9.38	noise
optimal, g=128	2.1	94.6	9.52	recognizable
optimal, g=32	2.5	94.6	**12.01**	near-baseline

Table 1: Sana 1.6B, three prompts, seed 0. The optimal scale is the decisive factor; group size provides the final polish. Weight reconstruction error for the optimal variants is 44.8% (g=128) and 43.2% (g=32).

Findings. (1) The scale rule is the dominant lever: swapping `absmean` for the optimal scale moves the model from noise to recognizable images at the same bit-width. (2) Smaller groups help monotonically but modestly. (3) Error feedback and per-row gain do not help and can hurt. (4) Keeping the six sensitive layers in higher precision is necessary; protecting additional whole blocks was not.

4 Activation-aware ternary: closing the gap to FP16

The data-free optimal scale minimizes the per-weight error $\|W - W_q\|_F$, but the network is sensitive to the per-output error $\|(W - W_q)X\|_F$ over its activations X. We close the remaining gap with two activation-aware components, calibrated on the model's *own generations* (six generic prompts; no external dataset).

GPTQ. We quantize the columns of W left to right; after quantizing column i with error e_i, we update the not-yet-quantized columns by $-e_i\, H^{-1}_{i,>i} / H^{-1}_{ii}$, where $H = \mathbb{E}[xx^\top] = \frac{1}{N}\sum xx^\top$ is

the input second moment, so the error is steered into directions the activations tolerate. The per-column quantizer is our group-wise optimal-scale ternary, with the group scale recomputed from the running weight at each group boundary.

Incoherence rotation. Outliers that a single group scale cannot capture are removed by a shared random orthogonal R. For each linear we quantize in the rotated basis,

$$y = Wx = (WR^\top)(Rx), \qquad R^\top R = I, \tag{6}$$

i.e. we store the quantized $WR^\top$ and apply $x \mapsto Rx$ at inference. A *single* R serves every same-width layer (the identity holds per layer), and the Hessian rotates analytically, $H \mapsto RHR^\top$, so no re-calibration is needed. The rotation mixes coordinates so that no weight is an outlier, making each group near-uniform in magnitude.

Method	data	eff. bits/wt	PSNR vs FP16 ↑	outcome
optimal scale, g=128	none	2.1	9.52	recognizable
optimal scale, g=32	none	2.5	12.01	near-baseline
+ GPTQ, g=128	self	2.1	14.91	good
+ GPTQ, g=32	self	2.5	16.53	strong
+ GPTQ + rotation, g=128	self	**2.1**	**17.01**	**near-FP16**

Table 2: Activation-aware ternary at the same 2.1-bit budget reaches near-FP16 quality. "self" = self-calibrated on the model's own generations.

5 Discussion and limitations

At ternary precision $\sim$43% of the weight magnitude is discarded, so outputs are stylized/posterized rather than pixel-identical; this is intrinsic to the bit-width, not the method. Our rotation uses a dense random orthogonal matrix for clarity; a deployment would replace it with a structured fast Hadamard transform ($O(d \log d)$) folded through the residual stream, removing the per-layer matmul overhead. We focus on a diffusion transformer here; both the data-free optimal scale and the activation-aware extension apply unchanged to decoder-only language models.

6 Conclusion

Data-free ternary quantization is far more capable than the standard `absmean` heuristic suggests. The closed-form MSE-optimal per-group scale— maximizing $(\sum_{\text{top-}k} |w|)^2/k$—turns a noise generator into a working 2-bit image model with no calibration data. The lesson is narrow but practical: *at extreme precision, spend your engineering on the scale.*

References

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Figure 1: Identical prompts and seed. Left: FP16 baseline. Middle: optimal-scale ternary, g=128 (2.1 bits/wt). Right: g=32 (2.5 bits/wt). Both are fully data-free with 94.6% of the transformer ternary. Contrast with `absmean`/error-feedback, which yield pure noise.

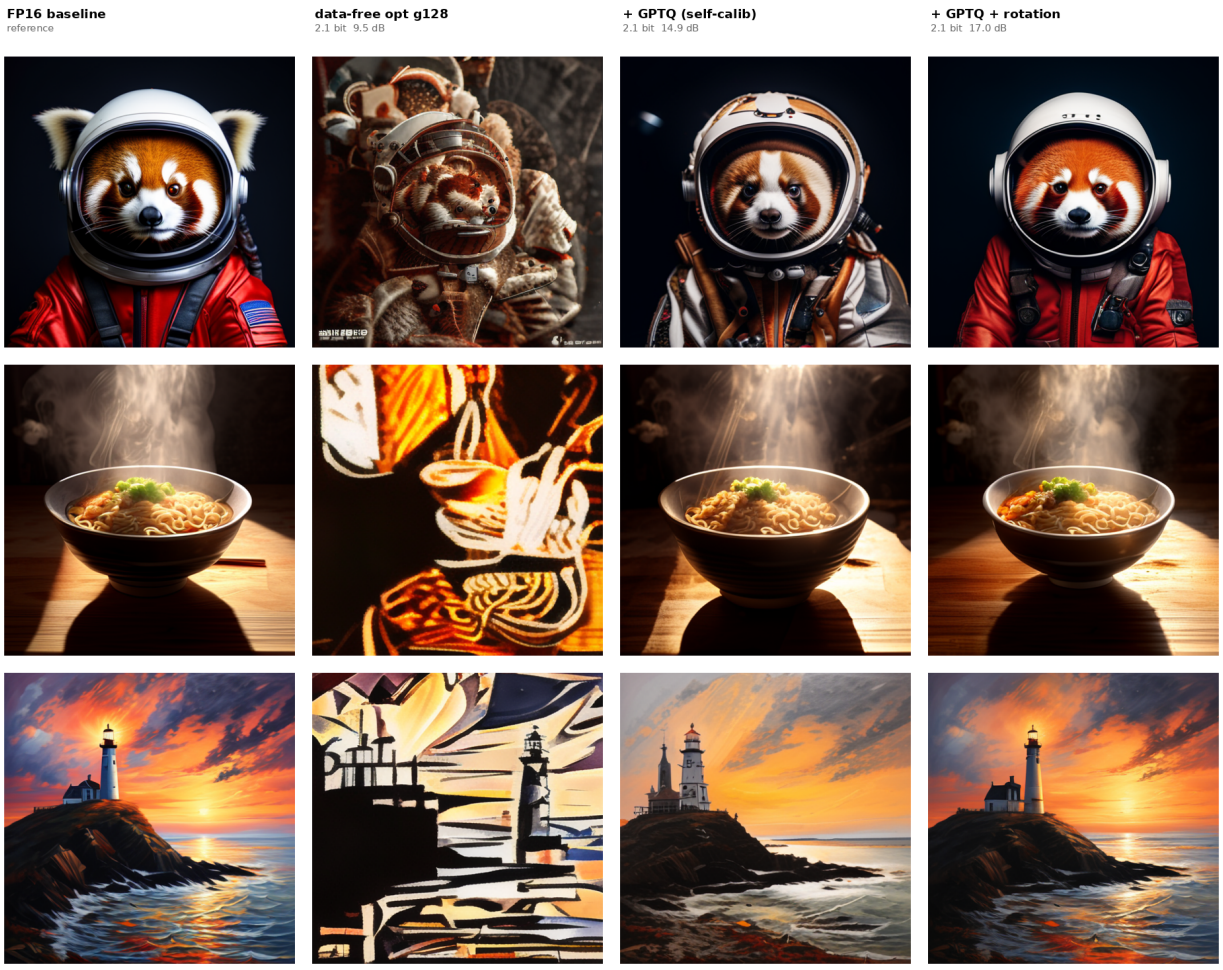


Figure 2: Same 2.1-bit budget (g=128, identical seed). Left to right: FP16 baseline; data-free optimal scale (9.5 dB, noise); +GPTQ self-calibration (14.9 dB); +incoherence rotation (17.0 dB, near-FP16).